

The Graph Neural Network Approach for Pravah

For building a predictive traffic management system, we propose a Graph Neural Network based representation of the urban road network using a directional graph composed of nodes and edges. The graph is designed to represent both the physical structure of the road network and the traffic-control conditions governing the movement of vehicles through it.

Our graph will contain road segments as nodes and junctions as edges, with traffic information assigned to each component as follows.

1. The Nodes

The nodes will represent individual **road segments** mapped from OpenStreetMap and associated with floating-car traffic data from sources such as TomTom. Each node will correspond to a distinct segment of the road network and will contain the properties required to describe its current traffic state and physical characteristics.

The properties of each segment node will include:

a. Segment ID

A unique identifier corresponding to the mapped road segment.

b. Average Speed

The current average vehicle speed observed on the segment, obtained from available traffic data.

c. Free-Flow Speed

The expected speed of the segment under uncongested traffic conditions. This provides a reference against which the current traffic state can be evaluated.

d. Congestion Ratio

A measure representing the current level of congestion on the segment relative to its free-flow condition.

e. Segment Length

The physical length of the corresponding road segment.

f. Traffic Pressure

A metric representing the traffic demand and imbalance associated with the segment, providing additional information about the pressure being exerted on the surrounding network.

Together, these features provide a representation of the current state of each road segment. The node representation can therefore be updated as new traffic observations become available and forms the primary state that the GNN is required to predict.

2. The Edges

The edges will represent the **junctions connecting adjacent road segments**. In addition to defining the connectivity of the road network, each edge will contain information about the traffic-control state and observations available at the corresponding junction.

The graph will be directional, allowing the direction of an edge to represent the permissible direction of traffic flow between connected road segments.

The properties of each junction edge will include:

a. Junction ID

A unique identifier corresponding to the junction connecting the respective road segments.

b. Traffic-Signal State

Information describing the current state of the traffic signal, including its active phase and relevant timing information. This allows the GNN to learn how the state of a junction influences the movement of traffic from one segment to another.

c. Camera-Derived Information

Information obtained from cameras positioned at or around the junction, including the number of heavy vehicles and the presence of emergency vehicles. Such information provides additional context regarding the composition and priority of traffic approaching or passing through the junction.

The edge therefore acts not only as a structural connection between two road segments, but also as the representation of the conditions governing traffic flow between them.

3. The Global Context

The GNN will additionally receive a global context representing factors that influence traffic across the network as a whole rather than a specific segment or junction.

This global context will include:

a. Time of Day

To capture recurring variations in traffic demand throughout the day.

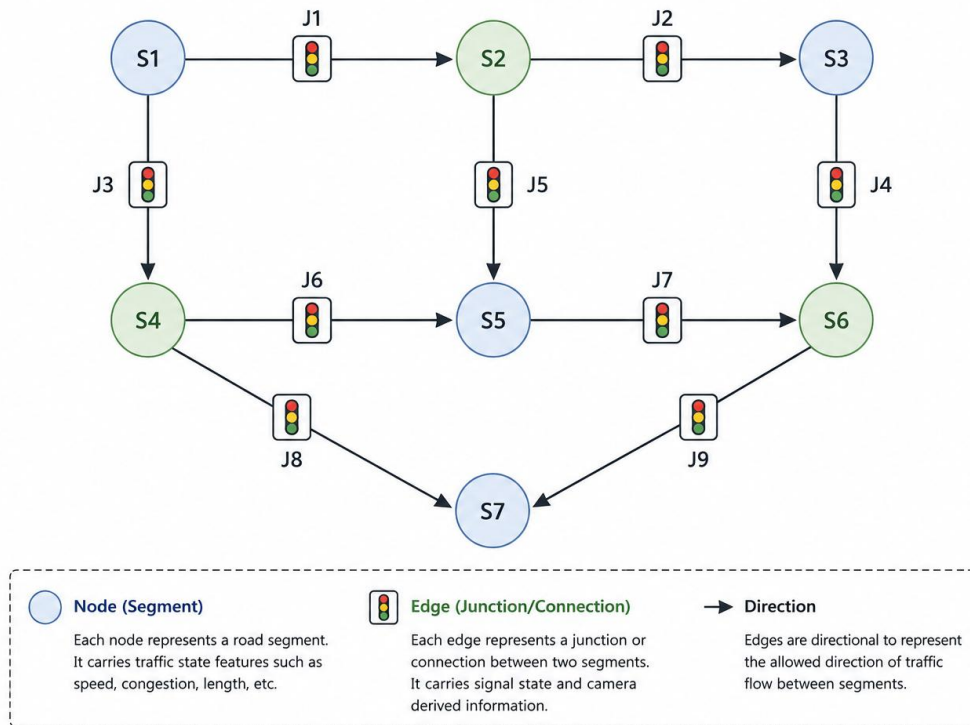
b. Day of Week

To account for systematic differences between weekdays, weekends, and other recurring traffic patterns.

c. Weather Conditions

To account for environmental conditions that may influence traffic behaviour across the network.

These variables will provide the GNN with the broader context required to interpret the state of individual nodes and edges.



The Aim of the GNN

The objective of the GNN will be to predict the future state of every road-segment node from the complete graph state observed at the current time.

Given the graph state at time (T), including the states of all road segments, junctions, traffic signals, camera-derived information, and global context, the model will predict the corresponding state of the network at $T + 60$ seconds:

The predicted state will then be linearly projected between the current observation and the 60-second forecast to derive a continuous estimate of traffic evolution throughout the intermediate period.

This approach is intended to address the latency associated with traffic data sources that are updated at approximately one-minute intervals. Rather than waiting for the next traffic observation to obtain an updated network state, PRAVAH can use the current observation to estimate how the network is expected to evolve over the next 60 seconds.

The Problem We Still Have

The primary challenge is the availability of data representing **different traffic-signal states and their corresponding effects on traffic flow**.

The available traffic data allows us to observe the traffic state of a road segment, but does not necessarily provide the synchronized signal state that produced that traffic condition.

Therefore, while we can observe that traffic changed, we cannot directly determine whether

that change was caused by a particular signal actuation, such as extending a green phase or increasing the duration of a red phase.

To train the system to understand this relationship, we require observations that associate a particular signal decision with the resulting traffic response. The fundamental question is therefore how to obtain or reconstruct this control-response relationship.

One possible approach is to use traffic simulation, apply different signal configurations, and observe the resulting changes in traffic flow. However, an important research question remains whether simulation is the only viable approach, or whether the impact of signal actuation can be inferred, at least partially, from real-world observational traffic data.